

# Elliott Schmechel

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## SUMMARY

Computer Science student and systems-focused builder. Ship end to end across Go, C++, TypeScript, and Python, from GPU training infrastructure and self-training AI agents to real-time edge apps. Multiple hackathon and challenge wins.

## EDUCATION

### Langara College

Associate of Science in Computer Science

Vancouver, BC

Sep. 2024 – Dec 2026

## EXPERIENCE

### Technical Content Engineer

Jun. 2026 – Aug. 2026

LicenseSpring (Cense Data Inc.)

Vancouver, BC

- Build reference sample applications and integration guides across the platform's SDKs (C++, Go, Python, .NET, Node.js) for developer onboarding
- Develop a demo sandbox environment and technical documentation for the licensing API and developer education platform
- Researching VM-detection robustness against minimal microVMs (Firecracker/KVM) to strengthen license-enforcement signatures

### Volunteer Fullstack Developer

Oct. 2025 – Present

United Nations Association – Vancouver

Hybrid

- Built and integrated 15+ responsive React components from Figma specs, improving page load speed by 25%
- Deployed on Cloudflare Workers/Pages with 99.9% uptime, reducing hosting costs by \$12/month

## PROJECTS

### Hermes Apprentice (Hermes Agent Challenge Winner, 1 of 4) | Go, Python, Unsloth, vLLM, ONNX

May 2026

- Cut recurring agent inference from multi-second upstream API calls to ~38ms p50 local latency by distilling matured task patterns into 18MB QLoRA specialists (Unsloth) served on vLLM with hot-swappable adapters
- Prevented quality regressions by gating every specialist behind a held-out F1 test (10-point delta vs. baseline) and a 5%-to-100% canary ramp that auto-demotes on agreement drop
- Automated pattern discovery from agent session logs with a Go observer, BGE-small ONNX embeddings, and HDBSCAN clustering, routing matches deterministically via cosine match in a Go proxy

### DataForAll (HTC 2026 Infrastructure Lead) | Python, FastAPI, Lambda Labs GPU, Kubernetes, Docker

Feb 2026

- Scaled a distributed training backend to 4 Kubernetes replicas with secrets management, container registry, and multi-region load balancing by writing the deployment manifests and provisioning H100 instances via the Lambda Labs API
- Reclaimed idle GPU spend (\$3.29/hr per H100) by building worker lifecycle management with heartbeat monitoring, stale-instance detection, and auto-cleanup, backed by S3 storage and PostgreSQL with Alembic migrations

### LearnLM (XHacks 2026, Best Use of SFUCoursesAPI) | TypeScript, Hono, Cloudflare Workers, React

Jan 2026

- Built the serverless backend on Cloudflare Workers, authoring 66% of commits across 6 D1 tables, 3-tier rate limiting, and a Vectorize RAG pipeline indexing 998 SFU courses
- Built a Model Context Protocol (MCP) server exposing 21 tools, integrated with VS Code, Cursor, and Claude Desktop
- Implemented real-time voice tutoring via WebSocket Durable Objects with Deepgram STT/TTS and Llama 3.1 8B at the edge

### Beeepd (JourneyHacks 2026, Lone Wanderer Award) | TypeScript, Hono, Drizzle ORM, D1, Leaflet

Jan 2026

- Solo-built a real-time proximity radar app in 12 hours, designing 6 Drizzle/D1 tables with indexed geospatial queries using the Haversine formula
- Implemented device-secret auth, friend request flows, and location updates with automatic 24-hour TTL expiry

## TECHNICAL SKILLS

**Languages:** Go, C++, Python, TypeScript, SQL, Bash, HTML/CSS

**Backend / Data:** REST APIs, Hono, FastAPI, Next.js, Drizzle ORM, PostgreSQL

**Infrastructure:** Docker, Kubernetes, Proxmox, Firecracker microVMs, Cloudflare Workers/Pages/D1, GitHub Actions, Lambda Labs

**ML / AI:** LoRA fine-tuning (Unsloth), local LLMs, MCP servers, RAG (Vectorize), Deepgram

**Linux / Systems:** Arch Linux (daily driver), Ubuntu, Debian, self-hosted services